

# **SCELBI-2106**

**Digital Group Video Card  
for the SCELBI Minicomputer**  
USER'S GUIDE

**Rev A PC Boards**

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**Revision History**

| Manual | PCB | Date           | Author      | Notes                                                                                |
|--------|-----|----------------|-------------|--------------------------------------------------------------------------------------|
| 1.01   | A   | 24 SEP<br>2026 | A.Kalinchuk | <ul style="list-style-type: none"><li>• Added amperage rating for inductor</li></ul> |

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## OVERVIEW

The SCELBI-2106 is a Digital Group (DG) video card adapted for the SCELBI minicomputer.

The card provides the following functions:

1. Accepts 8-bit data from a SCELBI output port and displays the corresponding character on a composite-video output.
2. Displays the 128-character MCM6571 set, addressed with seven bits.
3. Homes the write pointer to character position 0 when 377 octal (all ones) is written.

### Design Philosophy

The character-generation and timing circuitry is the original Digital Group design (Dr. Robert Suding). Additions on this PCB are:

- +5 V to +12 V and -9 V to -3.3 V conversion
- Latches for SCELBI output-port data
- Mechanical and power conventions used on other SCELBI boards (edge-connector footprint, +5 V and -9 V fuses)

Voltage conversion and data latching can be added to a stock DG card as a separate assembly. That arrangement requires additional interconnects between boards.

This design places those functions on one PCB. The card inserts in the original DG video-card connector and operates without further modification.

### Theory of Operation

Character generation is performed by an MCM6571 (128 characters). Seven 1101 SRAM devices store the 256-character frame, one device per bit.

The display is a 32-column by 8-row character grid (256 cells).

Each cell is written with two output instructions to the selected port. The first presents the character with bit 7 set (200 octal), which latches the data and blanks the raster. The second presents the same character with bit 7 cleared (AND 177 octal), which restores the raster. The blanking interval appears as a brief flash on each write.

The write pointer cannot be addressed randomly. It is reset to position 0 by writing 377 octal, then completing the cycle with 177 octal. Homing does not erase characters already stored in the frame buffer.

To clear the display, home the pointer and write 256 spaces (040 octal). Software that requires known column alignment should keep a column and row count in SCELBI RAM and pad with spaces to the next 32-character boundary. For

example, after 29 characters, three spaces advance to the next row when a line terminator is received.

# ASSEMBLY AND TEST INSTRUCTIONS

## General Assembly Tips

Install components in order of increasing height. When the board is inverted for soldering, the bench holds the parts against the PCB.

Component locations are shown on the PC Board Component Placement drawing (page 25).

Install all parts of one type (resistors, IC sockets, and so on) before soldering. Tape the parts lightly with low-adhesion tape (painter's tape), invert the board, and solder.

Seat DIP sockets fully against the PCB. Tack two diagonally opposite pins on each socket, then solder the remaining pins.

## Assembly

### Step 1 Install diodes and resistors

| Component         | Value         | Location                | Quantity |
|-------------------|---------------|-------------------------|----------|
| Diode             | 1N4742        | D1                      | 1        |
| Diode             | 1N914         | D2-D7                   | 6        |
| Diode, 6.2V Zener | 1N5341        | VR1                     | 1        |
| Diode, 12V Zener  | NTE5127       | VR2                     | 1        |
| Resistor, 1/4W    | 470 $\Omega$  | R1,R4,R14,R35           | 4        |
| Resistor, 1/4W    | 10K $\Omega$  | R2-R3,R5-R9,R24,R34     | 9        |
| Resistor, 1/4W    | 1K $\Omega$   | R10,R37                 | 2        |
| Resistor, 1/4W    | 1.3K $\Omega$ | R11                     | 1        |
| Resistor, 1/4W    | 2.2K $\Omega$ | R12,R16,R18-R23,R25-R33 | 17       |
| Resistor, 1/4W    | 220 $\Omega$  | R13                     | 1        |
| Resistor, 1/4W    | 22 $\Omega$   | R15                     | 1        |
| Resistor, 1/4W    | 22K $\Omega$  | R17                     | 1        |
| Resistor, 1/2W    | 10 $\Omega$   | R36                     | 1        |
| Resistor, 1/4W    | 15K $\Omega$  | R38                     | 1        |

## Step 2 Install DIP sockets

| Component                    | Location                             | Quantity |
|------------------------------|--------------------------------------|----------|
| 14-pin DIP socket, 0.3" wide | U1-U3,U13-U17,<br>U20-U21,U25,U34    | 12       |
| 16-pin DIP socket, 0.3" wide | U4-U5,U7-U12,U18,<br>U22-U24,U26-U33 | 21       |
| 24-pin DIP socket, 0.6" wide | U19                                  | 1        |

See General Assembly Tips. Tack each socket with two diagonally opposite pins. Complete soldering after all sockets are tacked.

## Step 3 Install ceramic capacitors

| Component          | Value    | Location   | Quantity |
|--------------------|----------|------------|----------|
| Capacitor, Mica    | 100 pF   | C1         | 1        |
| Capacitor, Ceramic | 0.001 uF | C2-C3,C20  | 3        |
| Capacitor, Ceramic | 0.01 uF  | C5,C18,C22 | 3        |
| Capacitor, Ceramic | 0.1 uF   | C6-C16     | 11       |

## Step 4 Install electrolytic capacitors

| Component                           | Value | Location   | Quantity |
|-------------------------------------|-------|------------|----------|
| Capacitor, Electrolytic, 25V, Axial | 10 uF | C4,C17,C21 | 3        |
| Capacitor, Electrolytic, 25V, Axial | 1 uF  | C19        | 1        |

Observe polarity. The positive lead is marked "+" on the PC Board Component Placement drawing.

## Step 5 Install transistors

| Component              | Value  | Location | Quantity |
|------------------------|--------|----------|----------|
| Transistor, NPN, TO-92 | 2N5129 | Q1       | 1        |
| Transistor, NPN, TO-92 | 2N2222 | Q2-Q3    | 2        |

Observe transistor orientation. Seat each device fully before soldering.

#### Step 6 Install inductor

| Component        | Value           | Location | Quantity |
|------------------|-----------------|----------|----------|
| Inductor, Radial | 391 uH, >300 mA | L1       | 1        |

#### Step 7 Install fuse holders

| Component   | Location       | Quantity |
|-------------|----------------|----------|
| Fuse holder | F1 (2), F2 (2) | 4        |

Fuse holders are polarized. Orient the retaining lip so that it prevents the fuse from sliding out of the holder. **Important: Ensure the fuse holders don't contact the passing traces below it.**

#### Step 8 Install crystal

| Component | Value     | Location | Quantity |
|-----------|-----------|----------|----------|
| Crystal   | 5.990 MHz | X1       | 1        |

A 5.990 MHz crystal may be difficult to obtain. A nearby frequency (for example 5.900 MHz) may be substituted; horizontal and vertical timing will change with crystal frequency.

#### Step 9 Install jumpers

Install the following jumpers between U23 and the edge-connector J1. Pin 23 is the bottom right-most pin and pins 24 and 25 follow it.

| From | To |
|------|----|
|------|----|

|            |           |
|------------|-----------|
| U23 Pin 5  | J1 Pin 23 |
| U23 Pin 12 | J1 Pin 24 |
| U23 Pin 13 | J1 Pin 25 |

#### Step 10 Install fuses

| Component     | Value              | Location | Quantity |
|---------------|--------------------|----------|----------|
| Fuse, 8AG, FB | $\frac{1}{2}$ A    | F1       | 1        |
| Fuse, 8AG, FB | 1- $\frac{1}{2}$ A | F2       | 1        |

#### Step 11 Inspection

1. Inspect all solder joints for opens and bridges.
2. Check for shorts between edge-connector pins 3 and 6 (GND and +5 V) and between pins 3 and 7 (GND and -9 V). Installed components produce a measurable resistance; a near-zero reading indicates a short. **Do not proceed until any short is cleared.**

#### Step 12 Clean the board

Water-soluble or "organic core" flux must be cleaned before ICs are inserted:

1. Wash the board under hot running water and scrub with a brush for several minutes. Do this after soldering is complete and before ICs are inserted.
2. Remove water with compressed air, including under components and inside IC sockets. A shop compressor is more effective than an aerosol duster.
3. Dry the board near a heat source for approximately 30 minutes.

Cleaning is optional with rosin-core solder. Use a chemical flux remover; a brush may be used to loosen residue. Inspect solder joints again after cleaning.

#### Step 13 Install ICs

Be careful not to fold pins under the body. Pin 1 faces the top of the board on all ICs. Straighten DIP leads by pressing each row against a flat surface before insertion.

IC locations and orientation are shown on the PC Board Component Placement drawing.

| Component | Value | Location | Quantity |
|-----------|-------|----------|----------|
|-----------|-------|----------|----------|

|                                                 |         |              |   |
|-------------------------------------------------|---------|--------------|---|
| IC, DIP, Dual 4-Input NAND Gate                 | 7420    | U1           | 1 |
| IC, DIP, Quad 2-Input NAND Gate                 | 7400    | U2-U3,U14    | 3 |
| IC, DIP, 4-Bit Binary Counter                   | 74193   | U4-U5,U7-U12 | 8 |
| IC, DIP, Quad 2:1 Multiplexer                   | 74157   | U6,U26       | 2 |
| IC, DIP, Quad 2-Input NAND Gate, Open Collector | 7401    | U13          | 1 |
| IC, DIP, Quad 2-Input AND Gate                  | 7408    | U15,U20      | 2 |
| IC, DIP, 8-Input NAND Gate                      | 7430    | U16,U34      | 2 |
| IC, DIP, Hex Inverter                           | 7404    | U17,U21      | 2 |
| IC, DIP, 8-Bit Shift Register                   | 74165   | U18          | 1 |
| IC, DIP, 8192-Bit Character Generator           | MCM6571 | U19          | 1 |
| IC, DIP, Dual Monostable Multivibrator          | 74123   | U22          | 1 |
| IC, DIP, Quad D Flip-Flop                       | 74175   | U23-U24      | 2 |
| IC, DIP, Triple 3-Input NAND Gate               | 7410    | U25          | 1 |
| IC, DIP, 256-Bit SRAM                           | 1101    | U27-U33      | 7 |

#### Step 14 Inspection

Inspect the assembly for the following defects:

- IC inserted backwards
- Bent-under IC pins
- Incorrect IC in any position
- ICs not fully seated

#### Testing

This procedure verifies basic operation of the SCELBI-2106. It is not a complete troubleshooting guide.

A working SCELBI and a connection from an output port to the card are required. Appendix A describes an enclosure and interconnect cables.

#### Step 1 Installation

Insert the SCELBI-2106 into the SCELBI Video Interface edge connector.

#### Step 2 Connections

Connect an I/O cable from SCELBI output port 17 (the last output port) to the 11-pin Amphenol connector on the SCELBI Video Interface.

Connect +5 V, -9 V, and ground between the SCELBI power supply and the corresponding binding posts on the SCELBI Video Interface.

Connect a composite-video monitor to V1.

### Step 3 Test video output

Apply power. The monitor should show a blank or weakly illuminated raster, or residual characters. Absence of a raster indicates a fault in the Video Interface or its power and video connections.

If a raster is present, enter the following program at memory address 000 000. The program writes "A" using the two-write protocol (bit 7 set, then bit 7 cleared) on output port 17:

| Address | Machine Code | Assembly Code |
|---------|--------------|---------------|
| 0       | 006 301      | LAI 301       |
| 2       | 137          | OUT 17        |
| 3       | 006 101      | LAI 101       |
| 5       | 137          | OUT 17        |
| 6       | 377          | HLT           |

Set the LO and HI address registers to 0 (066 000 and 056 000). For each byte, execute LMI (076), then INL (060) to advance the pointer.

Transfer control with JMP 000 000 (104 000 000). The processor executes the program and halts. The first character cell should display "A".

Display of "A" in the first cell indicates that the SCELBI-2106 write path is operating.

## APPENDIX A: BUILDING AN ENCLOSURE

This appendix is a construction outline for an enclosure. Dimensions, hole sizes, and hardware may be varied. The part numbers are examples, not mandatory substitutions.

### Step 1 Prepare enclosure

A suitable enclosure is 12 in × 10 in × 3 in (for example, BUD AC-413). Mark the front panel for the three binding posts and the composite-video connector, drill the holes, and fit insulating bushings so the binding posts do not contact the chassis. Hole diameters depend on the hardware used.

Punch the Amphenol connector opening with a Greenlee 732 punch, 1-11/64 in, or equivalent.

### Step 2 Add labels

Apply front-panel legends: P1, +5, GND, -9, and V1.

Add a title legend: SCELBI VIDEO INTERFACE.

Dry-transfer lettering (for example, Letraset) may be used. Adhesion on metal is often poor so a better alternative is recommended.

### Step 3 Mount edge connector

Mount the edge connector (for example, EDAC 305-044-520-202) approximately 2-½ in from the front panel, centered laterally. Mark and drill for approximately 1 in L-brackets (for example, Keystone 1570). Do not fasten the connector to the brackets until wiring is complete.

### Step 4 Mount power and data connectors

P1 is an Amphenol 86-CP11. Binding posts may be any type that fits the drilled holes (for example, Keystone 4093, red). The video connector is a chassis-mount RCA jack with solder terminals.

### Step 5 Wire up edge connector

Solder the edge-connector wiring according to the interconnection table below.

Wire the power pins to the binding posts with 16-20 AWG. Wire the video pins to the RCA jack. Wire the remaining pins to the Amphenol connector. Then mount the edge connector on the L-brackets.

| Edge Connector | To |
|----------------|----|
|----------------|----|

|    |                        |
|----|------------------------|
| 1  | V1 center pin          |
| 3  | GND, V1 outer pin/body |
| 6  | +5                     |
| 7  | -9                     |
| 14 | P1-9                   |
| 15 | P1-1                   |
| 16 | P1-2                   |
| 17 | P1-3                   |
| 18 | P1-4                   |
| 19 | P1-5                   |
| 23 | P1-6                   |
| 24 | P1-7                   |
| 25 | P1-8                   |

#### Step 6 Insert PCB

Insert the SCELBI-2106 in the edge connector. The board span is long enough to flex. Fit a spacer (for example, a wood block) at the opposite end, matching the connector height and spanning the enclosure width. Fasten the PCB to the spacer through the existing mounting holes.

#### Step 7 Finalize enclosure

Attach the bottom plate (for example, BUD BPA-1523) with at least four screws. Fit four rubber feet (for example, 3M SJ-5523).

#### Step 8 Connect to SCELBI and video

Connect the Video Interface to the SCELBI as follows.

| SCELBI Video Interface | To                 |
|------------------------|--------------------|
| P1                     | SCELBI OUT PORT 17 |
| +5                     | SCELBI PSU +5      |
| -9                     | SCELBI PSU -9      |
| GND                    | SCELBI PSU GND     |

|    |                         |
|----|-------------------------|
| V1 | Composite-Video Monitor |
|----|-------------------------|

I/O cables may be made from Belden 8747 12-conductor cable and Amphenol hardware (one 86-CP11, one 78-S11, two 86-3-24). The connectors have 11 contacts; leave one conductor unused.

## APPENDIX B: SOFTWARE

### CMON

Creed Monitor (CMON) relocated to page 076 and modified for keyboard input and SCELBI-2106 video output.

```
; Modified Creed Monitor for SCELBI + Digital Group video
; Relocated to page 076 (original SCELBI TTY ROM page)
; Keyboard: INP 004, ACK OUT 014
; Video: OUT 017 (two-write BIT7 strobe, 377 = home)
;
; RAM scratch (unchanged):
; 016-374 3-byte JMP constructed at runtime
; 016-377 one-byte save
;
; Start address: 076-000

ROMPAGE: EQU 076
RAMPAGE: EQU 016
JMPADDR: EQU 374
RSAVADR: EQU 377

CR: EQU 015
PROMPT: EQU 076
DASH: EQU 055
COLON: EQU 072
SPACE: EQU 040
ZERO: EQU 060
EIGHT: EQU 070

KEYIN: EQU 004
KEYOUT: EQU 014
OUTPORT: EQU 017

ORG ROMPAGE#0
START:
    LAI 103                ; 'C' - display HL and contents
CMD:
    CAL CLROUT              ; clear 256-char screen and home
    LHI \HB\TTBL
    LLI \LB\TTBL
CMD2:
    CPM
    JTZ CMDFND
    INL
    INL
    JFZ CMD2
ILLCMD:
NXTCMD:
    LAI SPACE
    CAL OUTPUT
    LAI PROMPT
    CAL OUTPUT
```

```

GETCHAR:
    INP KEYIN
    NDA
    JFS GETCHAR          ; wait until bit 7 set
    OUT KEYOUT           ; ack
    NDI 177
    CAL OUTPUT
    CPI EIGHT
    JFS CMD              ; not 0-7: treat as command
NUM:
    CPI ZERO
    JTS ILLCMD
    NDI 007
    LCA
    LAB
    RLC
    RLC
    RLC
    NDI 370
    ORC
    LBA
    JMP GETCHAR
EX:
    LAE
    LCD
    JMP JMPIND
CMDFND:
    INL
    LAM
    LCH
JMPIND:
    LLI \LB\JMPINSTR
    LHI \HB\JMPINSTR
    LMI 104
    INL
    LMA
    INL
    LMC
    JMP JMPINSTR
CX:
    JMP EX
CH:
    LDB
    JMP DHL
CL:
    LEB
    JMP DHL
CW:
    LHD
    LLE
    LMB
CI:
    INE
    JFZ DHL
    IND
    JMP DHL

```

```

CD:
    LAE
    DCE
    NDA
    JFZ DHL
    DCD

CC:
DHL:
    LAI SPACE
    CAL OUTPUT
    LAD
    CAL POCTAL
    LAI DASH
    CAL OUTPUT
    LAE
    CAL POCTAL
    LAI COLON
    CAL OUTPUT
    LHD
    LLE
    LAM
    CAL POCTAL
    JMP NXTCMD

POCTAL:
    LHI \HB\REGSAVE
    LLI \LB\REGSAVE
    LMA
    RLC
    RLC
    NDI 003
    CAL POCTAL2
    LLI \LB\REGSAVE
    LAM
    RAR
    RAR
    RAR
    CAL POCTAL2
    LLI \LB\REGSAVE
    LAM

POCTAL2:
    NDI 007
    ORI ZERO
    JMP OUTPUT

CLROUT:
    LCA
    LAI 377 ; home first (also writes 177 / white box)
    OUT OUTPUT
    LAI 177 ; drop BIT7 so the next write has a rising
edge
    OUT OUTPUT
    LHI 000

CONT:
    LAI 240 ; space with BIT7 - overwrites the home
glyph
    OUT OUTPUT
    NDI 177

```

```

        OUT  OUTPORT
        INH
        LAH
        CPI  000
        JFZ  CONT
        LAC
        RET
OUTPUT:
        LCA
        ORI  200
        OUT  OUTPORT
        NDI  177
        OUT  OUTPORT
        LAC
        RET

        ORG  ROMPAGE#362
TTBL:
        DATA  'L'
        DATA  \LB\CL
        DATA  'H'
        DATA  \LB\CH
        DATA  'I'
        DATA  \LB\CI
        DATA  'W'
        DATA  \LB\CW
        DATA  'X'
        DATA  \LB\CX
        DATA  'D'
        DATA  \LB\CD
        DATA  'C'
        DATA  \LB\CC

```

#### **P76-DG/KBD**

This software replaces the original page-076 I/O PROM (TTY or oscilloscope). It uses the SCELBI Keyboard Interface for input and the SCELBI-2106 for output when running the Monitor/Editor/Assembler (MEA).

```

;
; SCELBI page-76 I/O ROM
; Parallel keyboard + Digital Group 32x8 video for MEA
;
; Author: Artem Kalinchuk
;
        .ORG    76Q * 400Q
;
KBD:    EQU    4
ACK:    EQU    12
CRT:    EQU    15
COLS:   EQU    040Q
;

```

```

RCV:
    INP    KBD
    NDA
    JFS    RCV
    OUT    ACK
    LCA
    CAL    PRINT
    LAC
    RET

;
PRINT:
    LBA
    LEL
    NDI    177Q
    CPI    177Q
    JTZ    BSL
    CPI    076Q
    CTZ    GT1
    CPI    012Q
    JTZ    DONL
    CPI    040Q
    JTS    PX
    CAL    OUTCH

PX:
    LLE
    LAB
    LHI    0
    RET

DONL:
    CAL    PAD
    JMP    PX

;
PAD:
PAD1:
    LHI    1
    LLI    040Q
    LAM
    NDA
    RTZ
    LAI    240Q
    CAL    OUTCH
    JMP    PAD1

;
OUTCH:
    LHI    1
    CAL    PUT
    LLI    040Q
    LAM
    ADI    1
    LMA
    CPI    COLS
    RFZ
    XRA
    LMA
    LLI    041Q
    LAM

```

```

        ADI    1
        NDI    7
        LMA
        JTZ    HOME
        RET

;
HOME:
        LHI    1
        LAI    377Q
        OUT    CRT
        XRA
        OUT    CRT

FILL:
        LBA

CLR1:
        LAI    240Q
        OUT    CRT
        XRA
        OUT    CRT
        DCB
        JFZ    CLR1
        LLI    040Q
        LMB
        INL
        LMB
        RET

;
PAGE:
        LCI    2

PAGEL:
        CAL    HOME
        LMI    1
        INL
        LMC
        JMP    REPLAY

;
        .ORG    76Q * 400Q + 200Q
READENT:
P3CHK:
        LDH
        LHI    0
        LLI    127Q
        LAM
        CPI    002Q
        JFZ    RCV
        INH
        LLI    042Q
        LAM
        NDA
        JTZ    WAIT
        SUI    1
        LMA

SPRET:
        LAI    240Q
        LHD
        RET

```

```

;
        .ORG      76Q * 400Q + 232Q
ECHOENT:
ECHO:
        RET
BSL:
        LAI      134Q
;
        .ORG      76Q * 400Q + 235Q
WRITEENT:
        CPI      377Q
        RTZ
        JMP      PRINT
;
INIT:
        LDB
        LHI      0
        LLI      174Q
        LAM
        CPI      002Q
        JTZ      MPROMPT
        CPI      052Q
        JTZ      NL
        LLI      127Q
        LAM
        CPI      002Q
        LBD
        RTZ
DUMP:
        LCI      3
        CAL      PAGEL
NL:
        LLI      147Q
        LMI      006Q
        CAL      PAD
IDRET:
        LBD
        RET
MPROMPT:
NPROMPT:
        OUT      ACK
        LLI      127Q
        LAM
        LMI      000Q
        CPI      002Q
        CTZ      RCV
        JMP      HOME
WAIT:
WLP:
        CAL      RCV
        CPI      305Q
        JFZ      PAGE
GT1:
        LLI      127Q
        LMI      000Q
        RET

```

```

;
        .ORG    76Q * 400Q + 343Q
INITENT:
CLEARENT:
        JMP     INIT
PUT:
        ORI     200Q
        OUT     CRT
        XRA
        OUT     CRT
        RET
REPLAY:
        DCH
        LLI     307Q
        LAI     076Q
        LMA
        LBI     COLS
RP1:
        LAM
        CAL     PUT
        INL
        DCB
        JFZ     RP1
        INH
        RET
;
        .ORG    76Q * 400Q + 377Q
        DB      310Q
        END

```

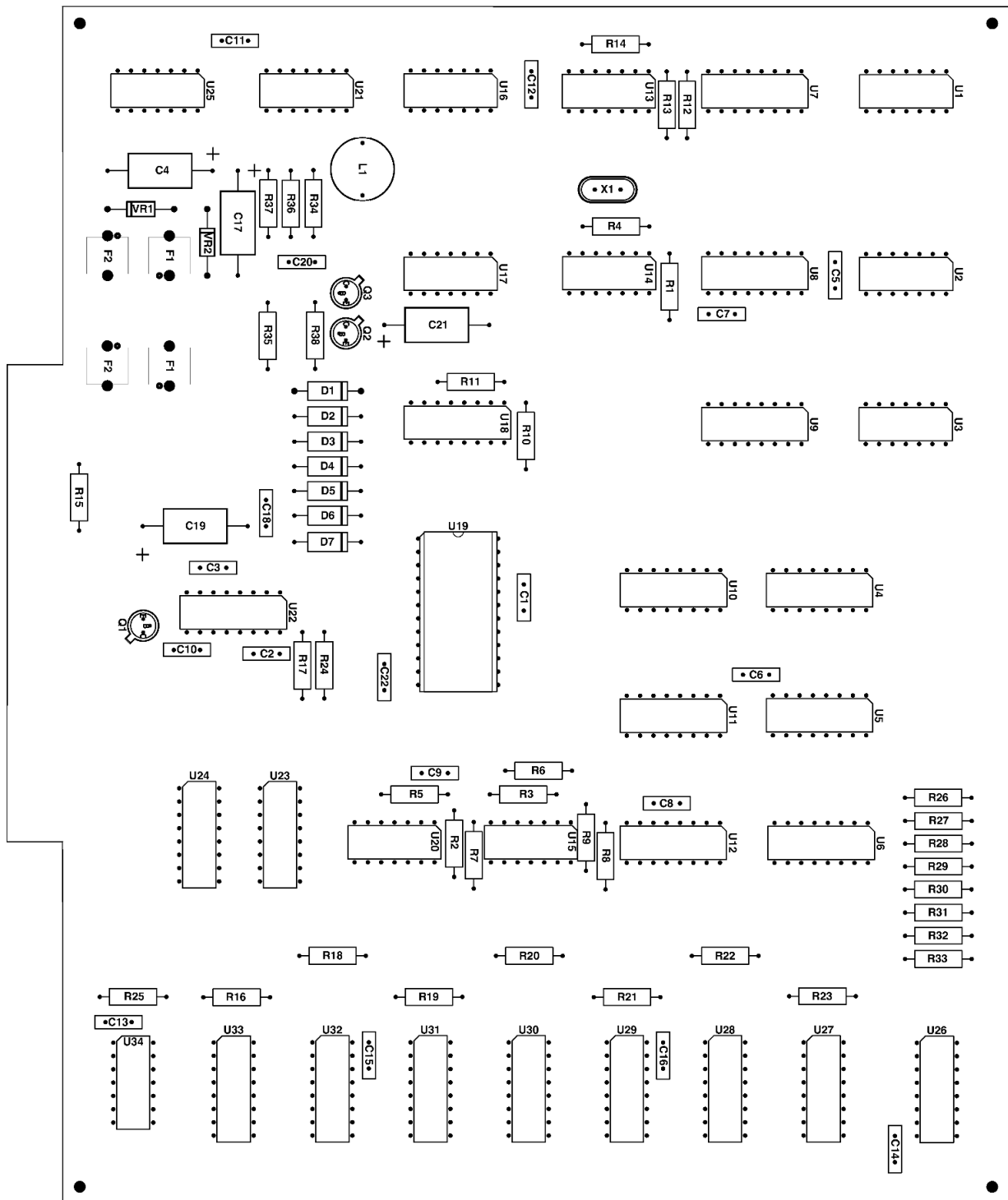
## PC BOARD COMPONENTS

| Component                    | Value         | Location                             | Quantity |
|------------------------------|---------------|--------------------------------------|----------|
| PC Board, SCELBI-2106, Rev A |               |                                      | 1        |
| Diode                        | 1N4742        | D1                                   | 1        |
| Diode                        | 1N914         | D2-D7                                | 6        |
| Diode, 6.2V Zener            | 1N5341        | VR1                                  | 1        |
| Diode, 12V Zener             | NTE5127       | VR2                                  | 1        |
| Resistor, 1/4W               | 470 $\Omega$  | R1,R4,R14,R35                        | 4        |
| Resistor, 1/4W               | 10K $\Omega$  | R2-R3,R5-R9,R24,R34                  | 9        |
| Resistor, 1/4W               | 1K $\Omega$   | R10,R37                              | 2        |
| Resistor, 1/4W               | 1.3K $\Omega$ | R11                                  | 1        |
| Resistor, 1/4W               | 2.2K $\Omega$ | R12,R16,R18-R23,<br>R25-R33          | 17       |
| Resistor, 1/4W               | 220 $\Omega$  | R13                                  | 1        |
| Resistor, 1/4W               | 22 $\Omega$   | R15                                  | 1        |
| Resistor, 1/4W               | 22K $\Omega$  | R17                                  | 1        |
| Resistor, 1/2W               | 10 $\Omega$   | R36                                  | 1        |
| Resistor, 1/4W               | 15K $\Omega$  | R38                                  | 1        |
| 14-pin DIP socket, 0.3" wide |               | U1-U3,U13-U17,<br>U20-U21,U25,U34    | 12       |
| 16-pin DIP socket, 0.3" wide |               | U4-U5,U7-U12,U18,<br>U22-U24,U26-U33 | 21       |
| 24-pin DIP socket, 0.6" wide |               | U19                                  | 1        |
| Capacitor, Mica              | 100 pF        | C1                                   | 1        |
| Capacitor, Ceramic           | 0.001 $\mu$ F | C2-C3,C20                            | 3        |
| Capacitor, Ceramic           | 0.01 $\mu$ F  | C5,C18,C22                           | 3        |
| Capacitor, Ceramic           | 0.1 $\mu$ F   | C6-C16                               | 11       |

|                                                 |                    |                |   |
|-------------------------------------------------|--------------------|----------------|---|
| Capacitor, Electrolytic, 25V, Axial             | 10 uF              | C4,C17,C21     | 3 |
| Capacitor, Electrolytic, 25V, Axial             | 1 uF               | C19            | 1 |
| Transistor, NPN, TO-92                          | 2N5129             | Q1             | 1 |
| Transistor, NPN, TO-92                          | 2N2222             | Q2-Q3          | 2 |
| Inductor, Radial                                | 391 uH,<br>>300 mA | L1             | 1 |
| Fuse holder                                     |                    | F1 (2) ,F2 (2) | 4 |
| Crystal                                         | 5.990<br>MHz       | X1             | 1 |
| IC, DIP, Dual 4-Input NAND Gate                 | 7420               | U1             | 1 |
| IC, DIP, Quad 2-Input NAND Gate                 | 7400               | U2-U3,U14      | 3 |
| IC, DIP, 4-Bit Binary Counter                   | 74193              | U4-U5,U7-U12   | 8 |
| IC, DIP, Quad 2:1 Multiplexer                   | 74157              | U6,U26         | 2 |
| IC, DIP, Quad 2-Input NAND Gate, Open Collector | 7401               | U13            | 1 |
| IC, DIP, Quad 2-Input AND Gate                  | 7408               | U15,U20        | 2 |
| IC, DIP, 8-Input NAND Gate                      | 7430               | U16,U34        | 2 |
| IC, DIP, Hex Inverter                           | 7404               | U17,U21        | 2 |
| IC, DIP, 8-Bit Shift Register                   | 74165              | U18            | 1 |
| IC, DIP, 8192-Bit Character Generator           | MCM6571            | U19            | 1 |
| IC, DIP, Dual Monostable Multivibrator          | 74123              | U22            | 1 |
| IC, DIP, Quad D Flip-Flop                       | 74175              | U23-U24        | 2 |
| IC, DIP, Triple 3-Input NAND Gate               | 7410               | U25            | 1 |
| IC, DIP, 256-Bit SRAM                           | 1101               | U27-U33        | 7 |
| Fuse, 8AG, FB                                   | $\frac{1}{2}$ A    | F1             | 1 |
| Fuse, 8AG, FB                                   | 1- $\frac{1}{2}$ A | F2             | 1 |

## PC BOARD DRAWINGS

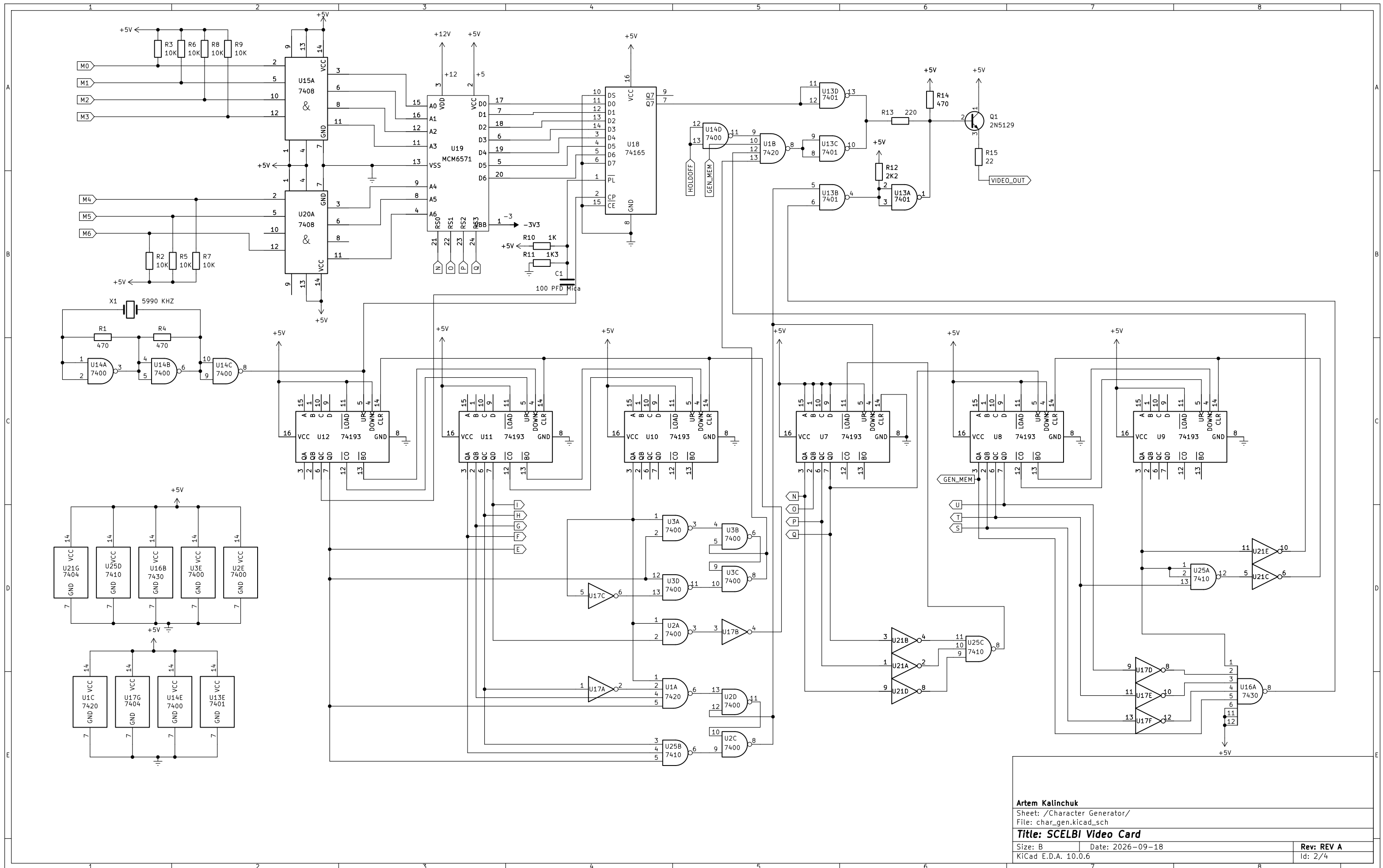
Component placement for Rev A is shown on the following pages.

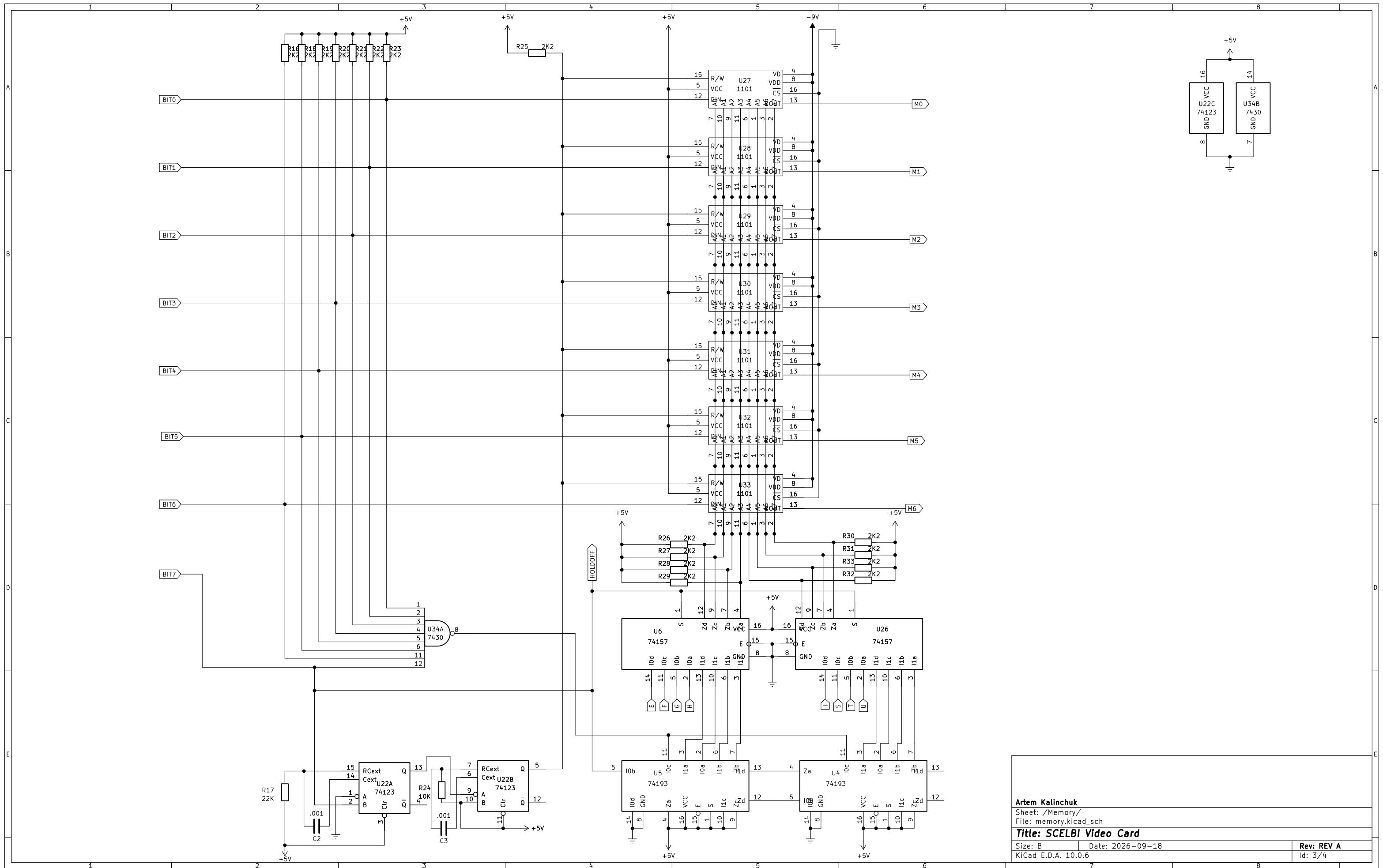


Rev A PC Board Component Placement

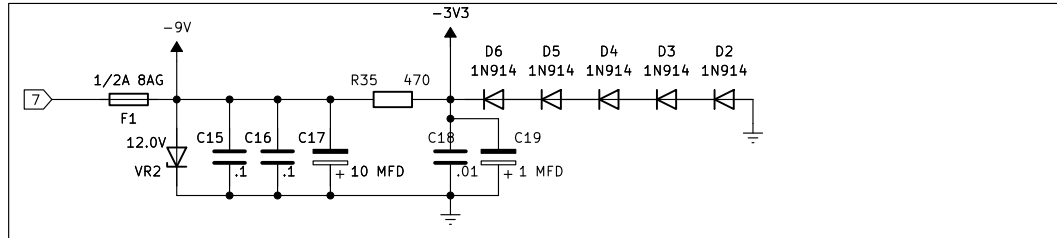
## SCHEMATIC DRAWINGS

Schematic diagrams for Rev A follow.

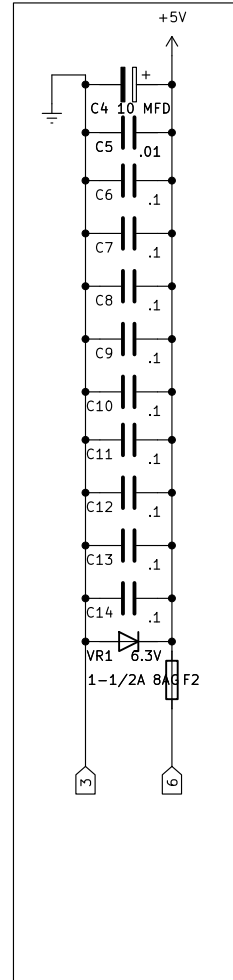




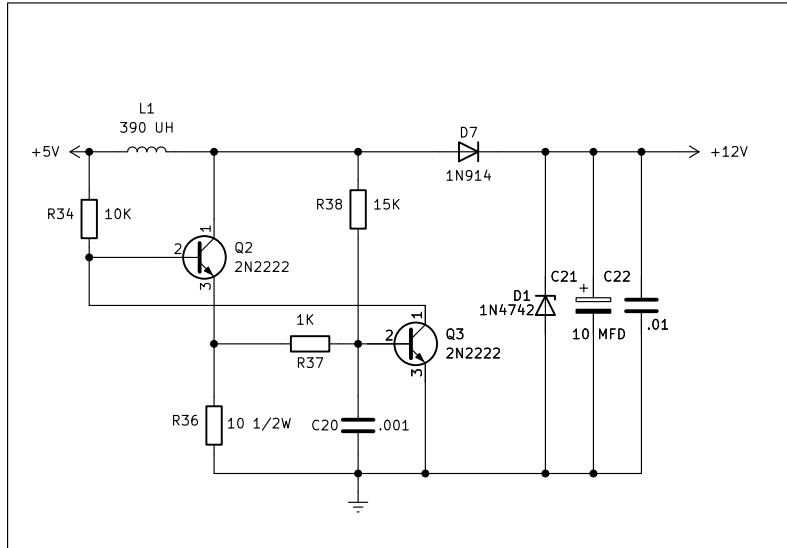
# -9V & -3V Power Conversion



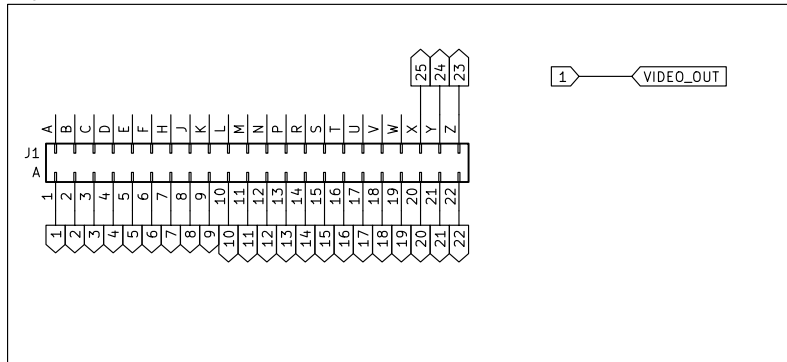
## +5V Power



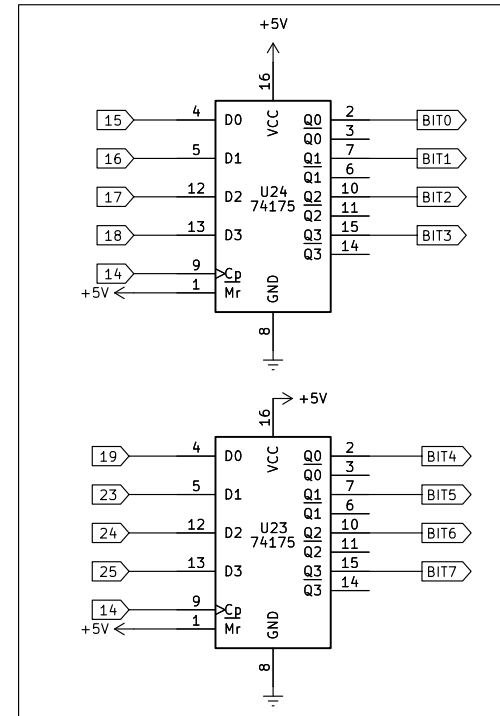
## +12V Power Conversion



## Edge Connector Connections



## Data Latch



Artem Kalinchuk

Sheet: /Connections & Power/  
File: connections\_power.kicad\_sch

**Title: SCELBI Video Card**

Size: A4 Date: 2026-09-18

KiCad E.D.A. 10.0.6

Rev: REV A

Id: 4/4